SeedHammer II — operator journey

Backing up a 5-of-12 multisig from files, end to end: constellation CLIs on the host, the firmware GUI in the emulator, and the plates that come out.

> **Test material only — never put funds behind these keys.**
> Every seed in this document was derived deterministically from the public string `"shibboleth journey cosigner <i>"` so the run reproduces exactly. They are public by construction.

What this run actually was

The *pathological* wallet from the constellation corpus: `wsh(multi(5, @0…@11))` — twelve cosigners, five signatures.

> **One correction, measured rather than assumed.**
> `CORPUS.md` calls its own C6 "pathological deeply-nested miniscript" entry *a placeholder*, noting the 8-nested-`or_d` form "actually fits single string (45 B)". The 12-key `multi(5,…)` policy used here comes instead from `phase-6-bucket-C.md`, which calls it "reliably >56 bytes bytecode" — no longer true of today's encoder. `md bytecode` reports **13 bytes**, and it encodes to a **single** md1 string: a keyless BIP-388 template does not pay per key. The chunking in this run comes from the **mk1 key cards**, which do carry xpubs — every one of the twelve splits in two.

Toolchain, as reported by the binaries

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md --version
md 0.13.0
[exit 0]

$ /scratch/code/shibboleth/mnemonic-key/target/release/mk --version
mk 0.12.1
[exit 0]

$ /scratch/code/shibboleth/mnemonic-secret/target/release/ms --version
ms 0.14.1
[exit 0]

$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me --version
me 0.5.1
[exit 0]

$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me-preview --version
me-preview 0.5.1
[exit 0]
```

The shape of the job

Policy	`wsh(multi(5,@0/<0;1>/*,…,@11/<0;1>/*))`
Cosigners	12, each at `m/48'/0'/0'/2'`, mainnet
md1 descriptor strings	1 (13-byte bytecode)
mk1 key-card strings	24 (12 keys × 2 chunks)
Public plates	25
Secret plates	1 (ms1 — typed on the machine, never over the wire)
Total plates	**26**

Twenty-six plates is the honest cost of a 5-of-12 held this way, and it is the number the operator most needs before starting.

The input files

Everything the operator supplies lives on disk before a single command runs. Contents below are complete and unedited.

THE WALLET POLICY

inputs/wallet-policy.txt

```
wsh(multi(5,@0/<0;1>/*,@1/<0;1>/*,@2/<0;1>/*,@3/<0;1>/*,@4/<0;1>/*,@5/<0;1>/*,@6/<0;1>/*,@7/<0;1>/*,@8/<0;1>/*,@9/<0;1>/*,@1
0/<0;1>/*,@11/<0;1>/*))
```

THE TWELVE SEEDS (SECRET — TEST MATERIAL)

inputs/seeds/cosigner-00.seed

```
ribbon relax cave lock library melody release ranch bird st
one fossil plate
```

inputs/seeds/cosigner-01.seed

```
more cotton early draw board party trap announce soft spend
farm output
```

inputs/seeds/cosigner-02.seed

```
output island tunnel permit ozone myth purse siren city mam
mal hobby owner
```

inputs/seeds/cosigner-03.seed

```
ketchup hobby brand attend cherry face shrimp route define
arch coconut annual
```

inputs/seeds/cosigner-04.seed

```
supreme twin piano juice oxygen audit various lounge paddle
flee speed divide
```

inputs/seeds/cosigner-05.seed

```
spoon eye blush hip grape supreme road blame chaos dwarf or
ange wool
```

inputs/seeds/cosigner-06.seed

```
cart inside barely depend domain lecture castle caution lat
in resource split coast
```

inputs/seeds/cosigner-07.seed

```
device accident orchard match coast pass upper decade snow
tumble cube radio
```

inputs/seeds/cosigner-08.seed

```
apart sustain chronic afraid thought science speed sting st
ay slim staff giggle
```

inputs/seeds/cosigner-09.seed

```
token quote smile section lazy slight problem people invest
network trophy swear
```

inputs/seeds/cosigner-10.seed

```
wrap category powder stadium flame oval hedgehog type blast
cricket chunk this
```

inputs/seeds/cosigner-11.seed

```
place cute page flash submit hard window scheme loyal cattl
e possible jealous
```

The input files, continued

THE TWELVE COSIGNER KEYS

inputs/keys/cosigner-00.xpub

```
# cosigner 0 — origin [ae6647ee/48h/0h/0h/2h]
xpub6F794Lv12fi3eLBDjDYxbbVk5GrvFQrqb8QpYRLbr8dNtmdtogc2o5K
iM3pLGd41phXxPw4ytcmwbDYmUfeRGuXE7CniQScVXrYnxz7AJE8
```

inputs/keys/cosigner-01.xpub

```
# cosigner 1 — origin [ff4bdd8b/48h/0h/0h/2h]
xpub6DyunbVQJAM2fHpJThHa24tAvbJTRZmbouQuUcZMdkrUvBHFo1hHANP
anDXPZfKSMcCxQMCqqgwnFJfjPDLPaqahwsgCMzeFsSyg5QFMU4G
```

inputs/keys/cosigner-02.xpub

```
# cosigner 2 — origin [b180e226/48h/0h/0h/2h]
xpub6EQfYuy9Hq9NwLNqhkZu4eAwoUdycVjeWRkVW2v2tZKaSgYZziFZdCr
6TqZJdpjeWiHA2REeJNcQNuHrj1bUot7hv2L3pnNCchmAKJi2Aar
```

inputs/keys/cosigner-03.xpub

```
# cosigner 3 — origin [7ab0a774/48h/0h/0h/2h]
xpub6EozX1gsda27873mLzSuK5n9nrUxJ4Z9LuNPRDkbQk8KG4riwQEjLjL
ybarddVedqavScZPhdbwgFA5MDRWkMUE7GYCJ2VSurq98gFu1NUm
```

inputs/keys/cosigner-04.xpub

```
# cosigner 4 — origin [bf3e6e44/48h/0h/0h/2h]
xpub6EnMa5NHmcfmAodnmaymYzxhVMp7Zg4sCdEbu9tDtcxgYE1RbXwXt1p
5S2iDtWbsBw6M6AFRgoQxxtWcSVSWkXHHMYfYcMiweUFPebdrYpz
```

inputs/keys/cosigner-05.xpub

```
# cosigner 5 — origin [c1826fbb/48h/0h/0h/2h]
xpub6EyaTpwePBqw9HFmqEHKPJLATVxGwvLaPgKgjY1s3EzBu7o9noT1Shm
i1wC4Y26xJV4N7adyjg4kyJybUUtYmR2e6cp8JBf7QPzzDhN9YiS
```

inputs/keys/cosigner-06.xpub

```
# cosigner 6 — origin [ecbd6d3c/48h/0h/0h/2h]
xpub6E7qf5omNGhrv4zEp46p7ZMY7NE5TkhiNAobBTfLZ7ZLSdmDKAnVSjh
fvj2kpCVqNyRsYLBZMHu3tfGTyzs8s2DGWhKcuocs3NJC1VC8xE3
```

inputs/keys/cosigner-07.xpub

```
# cosigner 7 — origin [7aa32b8d/48h/0h/0h/2h]
xpub6ENgKFaKvB7BjHcU7CQWRXn7DpHenFuPCe1sadzMC97Zfoa8uVBezdW
V9SfKJjWJwTNXzJuYQi5cjuDzr2QY5GGMJ4x6cez45r6BWC4RFjv
```

inputs/keys/cosigner-08.xpub

```
# cosigner 8 — origin [68c883f9/48h/0h/0h/2h]
xpub6Do4EgnUQcq9DGwXxircFtpE5NxAqGWYeJUSqbxsbfTDFsDvum2A8pj
A53F2jDosDs75jKiatutz9BVTgcvAto5EshMZdxPYUkip6M2YUos
```

inputs/keys/cosigner-09.xpub

```
# cosigner 9 — origin [5a3ae6a9/48h/0h/0h/2h]
xpub6FMrnRRrVL1FGGSiE4j9ve2SBSu1gf7CZTrkfeCn1mcQL96dN2URpwL
tQEsoZDMTU9Xk53rLEatEr4N7s367Q8mjXfC6s5X9guj4Pt8vZpi
```

inputs/keys/cosigner-10.xpub

```
# cosigner 10 — origin [ee9b5972/48h/0h/0h/2h]
xpub6DdB6ELpa6SXi6Jy4SENjXhG6AWGcJoXaULBAYscBbLkgWeVw1wQvTY
BWxqBoaPME4dXpJruhoBH39UkrhrSWkePhkwRGRYXx57CHSAGubn
```

inputs/keys/cosigner-11.xpub

```
# cosigner 11 — origin [dd7d1881/48h/0h/0h/2h]
xpub6EPABpiuEwhT2nA6YnHL7WnrmUudC2fWJzKgDaN8Gz5nBhedpLkaFty
VAkxeH9Bkam5k3dVxCXQJekKBr4LGVjirDneAMLjZv5eesz8Ft8k
```

Host step 1 — the descriptor

```
$ cat /tmp/claude-1000/-scratch-code-shibboleth-mnemonic-engrave/22fd28a4-d68a-47d6-82b1-8a8570fb5417/scratchpad/journey/inp
uts/wallet-policy.txt
wsh(multi(5,@0/<0;1>/*,@1/<0;1>/*,@2/<0;1>/*,@3/<0;1>/*,@4/<0;1>/*,@5/<0;1>/*,@6/<0;1>/*,@7/<0;1>/*,@8/<0;1>/*,@9/<0;1>/*,@1
0/<0;1>/*,@11/<0;1>/*))
[exit 0]

$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md encode --group-size 0 wsh(multi(5,@0/<0;1>/*,@1/<0;1>/*,@2/
<0;1>/*,@3/<0;1>/*,@4/<0;1>/*,@5/<0;1>/*,@6/<0;1>/*,@7/<0;1>/*,@8/<0;1>/*,@9/<0;1>/*,@10/<0;1>/*,@11/<0;1>/*))
md1ytpqqxpp3zcpydzk0zdt492xzr7r9qxfc
note: stdout is a keyless descriptor template (no keys)
[exit 0]

$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md bytecode md1ytpqqxpp3zcpydzk0zdt492xzr7r9qxfc
payload-bits: 100
payload-bytes: 13
hex: 22c200182188b0123456789ab0
note: stdout is a keyless descriptor template (no keys)
[exit 0]

$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md inspect md1ytpqqxpp3zcpydzk0zdt492xzr7r9qxfc
template: wsh(multi(5,@0/<0;1>/*,@1/<0;1>/*,@2/<0;1>/*,@3/<0;1>/*,@4/<0;1>/*,@5/<0;1>/*,@6/<0;1>/*,@7/<0;1>/*,@8/<0;1>/*,@9/
<0;1>/*,@10/<0;1>/*,@11/<0;1>/*))
n: 12
wallet-policy-mode: false
md1-encoding-id: 1c1c0feba95fef40d5a0e871f2463c52
wallet-descriptor-template-id: 726a666305756435b7c52c5b3fc69c41
wallet-policy-id: f05e8a1c282f7740bbfd902a759b5577
wallet-policy-id-fingerprint: 0xf05e8a1c
note: stdout is a keyless descriptor template (no keys)
[exit 0]
```

Host step 2 — a key card

One cosigner shown in full; the other eleven are identical in form.

```
$ cat /tmp/claude-1000/-scratch-code-shibboleth-mnemonic-engrave/22fd28a4-d68a-47d6-82b1-8a8570fb5417/scratchpad/journey/inp
uts/keys/cosigner-00.xpub
# cosigner 0 — origin [ae6647ee/48h/0h/0h/2h]
xpub6F794Lv12fi3eLBDjDYxbbVk5GrvFQrqb8QpYRLbr8dNtmdtogc2o5KiM3pLGd41phXxPw4ytcmwbDYmUfeRGuXE7CniQScVXrYnxz7AJE8
[exit 0]

$ /scratch/code/shibboleth/mnemonic-key/target/release/mk encode --xpub xpub6F794Lv12fi3eLBDjDYxbbVk5GrvFQrqb8QpYRLbr8dNtmdt
ogc2o5KiM3pLGd41phXxPw4ytcmwbDYmUfeRGuXE7CniQScVXrYnxz7AJE8 --origin-fingerprint ae6647ee --origin-path m/48'/0'/0'/2' --fro
m-md1 md1ytpqqxpp3zcpydzk0zdt492xzr7r9qxfc --group-size 0
mk1qpf7f8pqqsqhy6nxvwhxv3lwq5zg3vs76cp0whqh5fkzt3xcf63gqx2q8yj59pn36zmxs9tvvkpfpkahrevs9gswsqp6kz5gtp4sa5t5432d
mk1qpf7f8ppg9lsn2zdmpda6rg6cm06z68pesf3vlual5lym8g05e8cg79d06tf3jp2agyg8535cn7vl
note: stdout is watch-only — public keys only, cannot spend
[exit 0]

$ /scratch/code/shibboleth/mnemonic-key/target/release/mk inspect mk1qpmn4upqqsqhy6nxvwhxv3lwq5zg3vs76cp0whqh5fkzt3xcf63gqx2
q8yj59pn36zmxs9tvvkpfpkahrevs9gswsqp6kjy7ztspqdlh2j70 mk1qpmn4uppg9lsn2zdmpda6rg6cm06z68pesf3vlual5lym8g05e8cg79d06tf3jp2z2t
jad45qqhap
xpub:                xpub6F794Lv12fi3eLBDjDYxbbVk5GrvFQrqb8QpYRLbr8dNtmdtogc2o5KiM3pLGd41phXxPw4ytcmwbDYmUfeRGuXE7CniQScVXrY
nxz7AJE8
origin_fingerprint:  ae6647ee
xpub_fingerprint:    8791855f
origin_path:         48'/0'/0'/2'
  component[0]:       48h (hardened)
  component[1]:       0h (hardened)
  component[2]:       0h (hardened)
  component[3]:       2h (hardened)
policy_id_stubs:     726a6663
chunks:              2
  chunk[0]:           long (BCH variant)
  chunk[1]:           regular (BCH variant)
note: stdout is watch-only — public keys only, cannot spend
[exit 0]
```

ALL TWELVE, AS ENCODED

#	fingerprint	mk1 chunk 1/2	mk1 chunk 2/2
0	ae6647ee	mk1qpmn4upqqsqhy6nxvwhxv3lwq5zg3vs76cp0whqh5…	mk1qpmn4uppg9lsn2zdmpda6rg6cm06z68pesf3vlual…
1	ff4bdd8b	mk1qp2rwcpqqsqhy6nxv0l5hhvtq5zg3vs785q05fj45…	mk1qp2rwcppf85yqctkaapxmnt07mjglr6zd7lkh5e7j…
2	b180e226	mk1qp4ws3pqqsqhy6nxvwccpc3xq5zg3vs7wufpu05k6…	mk1qp4ws3ppm425ka4j8hp9h6nwzs5z47l0z3rhwnr3r…
3	7ab0a774	mk1qpuu24pqqsqhy6nxvdatpfm5q5zg3vs74h9qvgx6p…	mk1qpuu24pprtt7xm7r5n8lejuk9a3um0arv3te69v8v…
4	bf3e6e44	mk1qpextwpqqsqhy6nxvwlnumjyq5zg3vs748e0mw2c5…	mk1qpextwppqc64lyvss0trqzzah3jp0fc68g7rqu6d7…
5	c1826fbb	mk1qp0vxqpqqsqhy6nxv0qcymamq5zg3vs7c3zlcxs6p…	mk1qp0vxqpprggm8xltzc52l03jwc3n4c7f54ehyghvx…
6	ecbd6d3c	mk1qpw2mqpqqsqhy6nxv0kt6mfuq5zg3vs7f7drhpl27…	mk1qpw2mqppe7pvwaksnp3gy5unqvgm32amwm8983003…
7	7aa32b8d	mk1qp727jpqqsqhy6nxvda2x2udq5zg3vs7wf5s0sfdg…	mk1qp727jppqdwh7xccvj00tcwywaznfznv05c9lk5hl…
8	68c883f9	mk1qpj7wmpqqsqhy6nxvd5v3qleq5zg3vs7yw909j7cq…	mk1qpj7wmppvm7vuqdr6rxc6c9c97a7kcaj2jk7tmzgf…
9	5a3ae6a9	mk1qpcpp0pqqsqhy6nxvddr4e4fq5zg3vs7lzyle54xg…	mk1qpcpp0ppgcd6hyffmrlt98m2kxa79pdh5uw3y75wy…
10	ee9b5972	mk1qpr7lepqqsqhy6nxv0hfkktjq5zg3vs7p3wwdzd8h…	mk1qpr7leppem223hve8z8w6epwepftrn50zekhzdum0…
11	dd7d1881	mk1qpalkmpqqsqhy6nxv0wh6xypq5zg3vs7wwyena47t…	mk1qpalkmppmesaxh7fy50mx7ms70xd0xf6qpd0ghydl…

Host step 3 — the seed, and the line the tool will not cross

```
$ cat /tmp/claude-1000/-scratch-code-shibboleth-mnemonic-engrave/22fd28a4-d68a-47d6-82b1-8a8570fb5417/scratchpad/journey/inp
uts/seeds/cosigner-00.seed
ribbon relax cave lock library melody release ranch bird stone fossil plate
[exit 0]

$ /scratch/code/shibboleth/mnemonic-secret/target/release/ms encode --phrase ribbon relax cave lock library melody release r
anch bird stone fossil plate
ms10e ntrsq zu3dg yngxu pz9fd fkxpd xkfdl fsa39 6kgte j06n7
word count: 12
language: english (BIP-39 checksum valid)
passphrase: not stored in ms1 (record separately if used)
warning: stdout carries private key material (can spend) — redirect or encrypt (e.g. '> file.txt' or '| age -e ...')
[exit 0]
```

ms1 is refused on the radio path, by design.

Feeding the same seed's ms1 string to the NFC converter does not produce a tag:

```
$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me --in /dev/fd/63 --hex
me: refusing to emit ms1 over NFC: ms1 is secret seed entropy and must never be transmitted by radio (RF eavesdropping ris
k). Enter it by hand on the device: New > Input Seed > CODEX32.
[exit 3]
```

The descriptor, being public, converts without complaint:

```
$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me --in /dev/fd/63 --hex
0329d1012554006d64317974707171787070337a637079647a6b307a6474343932787a7237723971786663fe
[exit 0]
```

Host step 4 — validate the bundle, get the checklist

```
$ head -3 /tmp/claude-1000/-scratch-code-shibboleth-mnemonic-engrave/22fd28a4-d68a-47d6-82b1-8a8570fb5417/scratchpad/journe
y/inputs/backup-strings.txt
md1ytpqqxpp3zcpydzk0zdt492xzr7r9qxfc
mk1qpmn4upqqsqhy6nxvwhxv3lwq5zg3vs76cp0whqh5fkzt3xcf63gqx2q8yj59pn36zmxs9tvvkpfpkahrevs9gswsqp6kjy7ztspqdlh2j70
mk1qpmn4uppg9lsn2zdmpda6rg6cm06z68pesf3vlual5lym8g05e8cg79d06tf3jp2z2tjad45qqhap
[exit 0]

$ wc -l /tmp/claude-1000/-scratch-code-shibboleth-mnemonic-engrave/22fd28a4-d68a-47d6-82b1-8a8570fb5417/scratchpad/journey/i
nputs/backup-strings.txt
25 /tmp/claude-1000/-scratch-code-shibboleth-mnemonic-engrave/22fd28a4-d68a-47d6-82b1-8a8570fb5417/scratchpad/journey/input
s/backup-strings.txt
[exit 0]

$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me bundle --in /tmp/claude-1000/-scratch-code-shibboleth-mnemonic
-engrave/22fd28a4-d68a-47d6-82b1-8a8570fb5417/scratchpad/journey/inputs/backup-strings.txt --preview /tmp/claude-1000/-scrat
ch-code-shibboleth-mnemonic-engrave/22fd28a4-d68a-47d6-82b1-8a8570fb5417/scratchpad/journey/out/plates --png --manifest /tm
p/claude-1000/-scratch-code-shibboleth-mnemonic-engrave/22fd28a4-d68a-47d6-82b1-8a8570fb5417/scratchpad/journey/out/manifes
t.json
me: rendered plate 1 → /tmp/claude-1000/-scratch-code-shibboleth-mnemonic-engrave/22fd28a4-d68a-47d6-82b1-8a8570fb5417/scrat
chpad/journey/out/plates/plate-1.png
  … (plates 2–25 rendered) …
```

THE PLATE CHECKLIST, VERBATIM

```
me: backup needs 26 plates (25 public + ms1 on device):
  plate 1/26  md1 policy  → push via NFC & engrave
  plate 2/26  mk1 chunk 1/2  → push via NFC & engrave
  plate 3/26  mk1 chunk 2/2  → push via NFC & engrave
  plate 4/26  mk1 chunk 1/2  → push via NFC & engrave
  plate 5/26  mk1 chunk 2/2  → push via NFC & engrave
  plate 6/26  mk1 chunk 1/2  → push via NFC & engrave
  plate 7/26  mk1 chunk 2/2  → push via NFC & engrave
  plate 8/26  mk1 chunk 1/2  → push via NFC & engrave
  plate 9/26  mk1 chunk 2/2  → push via NFC & engrave
  plate 10/26  mk1 chunk 1/2  → push via NFC & engrave
  plate 11/26  mk1 chunk 2/2  → push via NFC & engrave
  plate 12/26  mk1 chunk 1/2  → push via NFC & engrave
  plate 13/26  mk1 chunk 2/2  → push via NFC & engrave
  plate 14/26  mk1 chunk 1/2  → push via NFC & engrave
  plate 15/26  mk1 chunk 2/2  → push via NFC & engrave
  plate 16/26  mk1 chunk 1/2  → push via NFC & engrave
  plate 17/26  mk1 chunk 2/2  → push via NFC & engrave
  plate 18/26  mk1 chunk 1/2  → push via NFC & engrave
  plate 19/26  mk1 chunk 2/2  → push via NFC & engrave
  plate 20/26  mk1 chunk 1/2  → push via NFC & engrave
  plate 21/26  mk1 chunk 2/2  → push via NFC & engrave
  plate 22/26  mk1 chunk 1/2  → push via NFC & engrave
  plate 23/26  mk1 chunk 2/2  → push via NFC & engrave
  plate 24/26  mk1 chunk 1/2  → push via NFC & engrave
  plate 25/26  mk1 chunk 2/2  → push via NFC & engrave
  plate 26/26  ms1 secret  → TYPE ON DEVICE (New > Input Seed > CODEX32); never via this tool
```

Host step 5 — a systemwide payload

The public cards can also be handed to the machine as one flash payload at `0x10D00000`. The digest is what the operator compares on screen.

```
$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me sysw pack --no-passphrase md1ytpqqxpp3zcpydzk0zdt492xzr7r9qxfc
--out /tmp/claude-1000/-scratch-code-shibboleth-mnemonic-engrave/22fd28a4-d68a-47d6-82b1-8a8570fb5417/scratchpad/journey/ou
t/sysw-public.bin
strength: no passphrase — BELOW the threshold
digest:   05f3 a2c0 7528 fbf5 03be b78c 2edf 524c
[exit 0]

$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me sysw show /tmp/claude-1000/-scratch-code-shibboleth-mnemonic-e
ngrave/22fd28a4-d68a-47d6-82b1-8a8570fb5417/scratchpad/journey/out/sysw-public.bin
sealed:   false
pub_len:  36
ct_len:   0
identity: e1af502140b8da818dfb69c5fe380bcb239505fbd0fef621dc967e3eb677fb27
digest:   05f3 a2c0 7528 fbf5 03be b78c 2edf 524c
[exit 0]

$ /scratch/code/shibboleth/mnemonic-engrave/target/release/me sysw wipe --fill zeros --out /dev/null
[exit 0]
```

The 25 public plates

Rendered by the `me-preview` sidecar, which reuses upstream SeedHammer curve math — this is the layout the machine will cut.

md1ytpqqxpp3zcpydzk0zdt492
xzr7r9qxfc

plate 1 — md1 policy

mk1qpr7lepqqsqhy6nxv0hfkkt
jq5zg3vs7p3wwdzd8hzkpwz902
d9kc5t2eft5hteqp
p0tj8yacjs57essr
vy0hxp6t5pn29vuk
wmhcs8v9lmw

plate 2 — key 0 [ae6647ee] chunk 1/2

mk1qpr7leppem223hve8z8w6ep
wepftrn50zekhzdum0xadvukdl
2w5xcwergvxzgsnf0y
rkxspee9gx

plate 3 — key 0 [ae6647ee] chunk 2/2

mk1qp2rwcpqqsqhy6nxv0l5hhv
tq5zg3vs785q05fj45mfq8sumu
57v5jergsejg9t3f
6he476ws0gxzx7x4
gzlzcpn7upqjafz3
f56alck3hwg

plate 4 — key 1 [ff4bdd8b] chunk 1/2

mk1qp2rwcppf85yqctkaapxmnt
07mjglr6zd7lkh5e7jvktz2jpx
qszmx3a92594l2d7m8
ctdmxr5r92

plate 5 — key 1 [ff4bdd8b] chunk 2/2

mk1qpw2mqpqqsqhy6nxv0kt6mf
uq5zg3vs7f7drhpl27xtwc3x0h
gs2qyrwl2htd0ydd
2m8numm0e8vlfvfe
dn4s3xgfspqjc0tq
sgz7ggysp67

plate 6 — key 2 [b180e226] chunk 1/2

mk1qpw2mqppe7pvwaksnp3gy5u
nqvgm32amwm89830033mvzkjq2
k9pad28ytx0p296v80
3dm09l3us0

plate 7 — key 2 [b180e226] chunk 2/2

mk1qp0vxqpqqsqhy6nxv0qcyma
mq5zg3vs7c3zlcxs6p0uns78cx
de6v2uw3duvrdqxn
egu75utyyk4vmypz
pq6n9vkwypcwz4mw
6rw70q5nw44

plate 8 — key 3 [7ab0a774] chunk 1/2

mk1qp0vxqpprggm8xltzc52l03
jwc3n4c7f54ehyghvx8f66egfn
kunvtj7d4sly9qjs80
jh6r5q8gxx

plate 9 — key 3 [7ab0a774] chunk 2/2

mk1qpj7wmpqqsqhy6nxvd5v3ql
eq5zg3vs7yw909j7cqa0un6pd4
x47rxpgnzt3eyvgk
5x3xyj698ym04tw0
ua7ua6a85ps5q472
fwu7494nrlq

plate 10 — key 4 [bf3e6e44] chunk 1/2

mk1qpj7wmppvm7vuqdr6rxc6c9
c97a7kcaj2jk7tmzgfrsxut0kn
nptpjhnnm2xr6anjza
x7wspvcvj7

plate 11 — key 4 [bf3e6e44] chunk 2/2

mk1qp4ws3pqqsqhy6nxvwccpc3
xq5zg3vs7wufpu05k62zp3z2yg
nayvvgm7f08maf30
9r66xkv6y2554yuc
4z5dx5ddsp26ewnn
x8gm3qsl2wd

plate 12 — key 5 [c1826fbb] chunk 1/2

The 25 public plates, continued

```
mk1qp4ws3ppm425ka4j8hp9h6n
wzs5z47l0z3rhwnr3rzfmr6lwe
vt8gxwfkrartt5uja7
srruflxzcp
```

plate 13 — key 5 [c1826fbb] chunk 2/2

```
mk1qpcpp0pqqsqhy6nxvddr4e4
fq5zg3vs7lzyle54xgxknnp2a6
78ccflnu0a92uwpe
tyag4yh3zyg60xga
7ztgjmpecp759yjg
jpvmj24xpc7
```

plate 14 — key 6 [ecbd6d3c] chunk 1/2

```
mk1qpcpp0ppgcd6hyffmrlt98m
2kxa79pdh5uw3y75wy4n0dsqyr
y4re3zh70wygh30ncv
25zx6877lh
```

plate 15 — key 6 [ecbd6d3c] chunk 2/2

```
mk1qpextwpqqsqhy6nxvwlnumj
yq5zg3vs748e0mw2c5nqufu2yx
tlt3p73lle7xxyxu
xgl528kntx50svyx
z2wcu674cp42rxe6
y9r9m9u5zml
```

plate 16 — key 7 [7aa32b8d] chunk 1/2

```
mk1qpextwppqc64lyvss0trqzz
ah3jp0fc68g7rqu6d7ntvk90av
8477mlqm8xsxtc8kw7
la42r6tdme
```

plate 17 — key 7 [7aa32b8d] chunk 2/2

```
mk1qpmn4upqqsqhy6nxvwhxv3l
wq5zg3vs76cp0whqh5fkzt3xcf
63gqx2q8yj59pn36
zmxs9tvvkpfpkahr
evs9gswsqp6kjy7z
tspqdlh2j70
```

plate 18 — key 8 [68c883f9] chunk 1/2

```
mk1qpmn4uppg9lsn2zdmpda6rg
6cm06z68pesf3vlual5lym8g05
e8cg79d06tf3jp2z2t
jad45qqhap
```

plate 19 — key 8 [68c883f9] chunk 2/2

```
mk1qpuu24pqqsqhy6nxvdatpfm
5q5zg3vs74h9qvgx6p5tcelvu5
ys79h3l6k6p077rf
p0w7gc6wpgs276z0
28zcp7thvplyxhte
w8rqed34etp
```

plate 20 — key 9 [5a3ae6a9] chunk 1/2

```
mk1qpuu24pprtt7xm7r5n8leju
k9a3um0arv3te69v8vvzzha94k
h8yunz4k2za6scpqc7
ks798th9d7
```

plate 21 — key 9 [5a3ae6a9] chunk 2/2

```
mk1qpalkmpqqsqhy6nxv0wh6xy
pq5zg3vs7wwyena47twxpez7ws
m3c4x6qus3sjhqrq
sv6uczrk2tjugkf8
c8xe5p5pgpgvwt75
h39vlurnemn
```

plate 22 — key 10 [ee9b5972] chunk 1/2

```
mk1qpalkmppmesaxh7fy50mx7m
s70xd0xf6qpd0ghydlfs4knp2k
4pvspusczvuj59rq63
s28ntuf6n6
```

plate 23 — key 10 [ee9b5972] chunk 2/2

```
mk1qp727jpqqsqhy6nxvda2x2u
dq5zg3vs7wf5s0sfdgl3jpqafg
mt0kwplhvgsc3zzx
fzj6ydsrdc80lagg
svc9skp85pgg8ct3
d2p465r0z36
```

plate 24 — key 11 [dd7d1881] chunk 1/2

```
mk1qp727jppqdwh7xccvj00tcw
ywaznfznv05c9lk5hlmr6p0eqh
ty79q0ust5ft6z03qu
l6vu625jhr
```

plate 25 — key 11 [dd7d1881] chunk 2/2

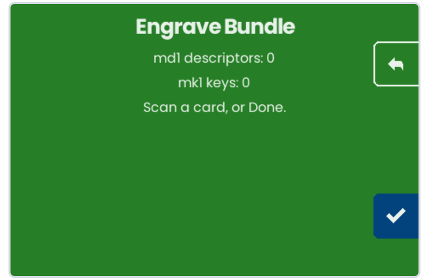
> **Plate 26 is not here, and cannot be.**
>
> The seed plate is engraved from an ms1 string typed on the machine. No host tool in this constellation will render, transmit or preview it.

On the machine — finding the program

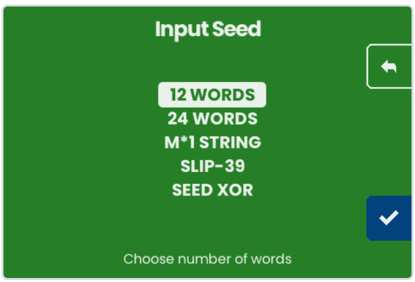
The emulator runs the real firmware GUI compiled to wasm, driven only through the touch input the machine itself has.

Boot: the program carousel, 8 entries

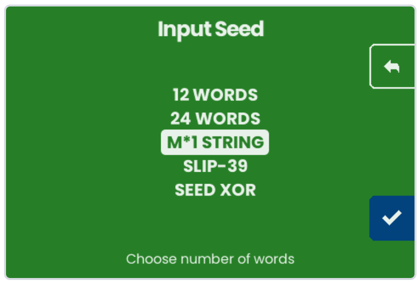
Engrave Bundle — takes md1 + mk1 cards



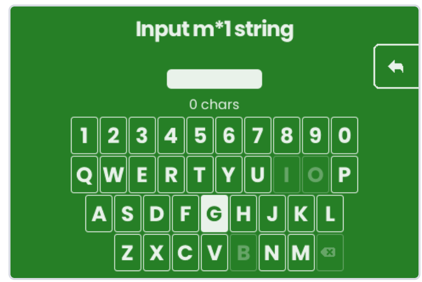
The gatherer, waiting for cards



Backup Wallet: seed source first



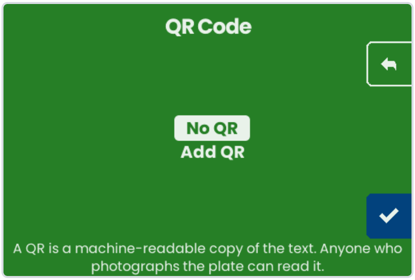
M*1 STRING selected



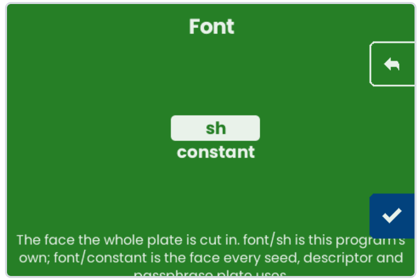
bech32 keyboard — I, O and B are absent

The eight programs, read off the carousel:
Backup Wallet · BIP-39 Password · Engrave Text · Account Xpub · Engrave Bundle · Engrave Single-Sig · Engrave Multisig · BIP-85 Child Seed.

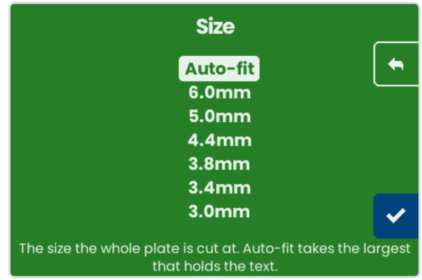
On the machine — composing a plate



QR: the warning is that a photo of the plate is a copy of the text



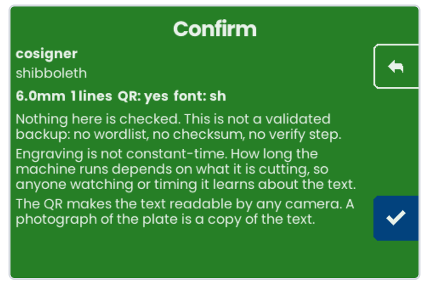
Font: font/sh, or the face every seed and descriptor plate uses



Size: auto-fit takes the largest that holds

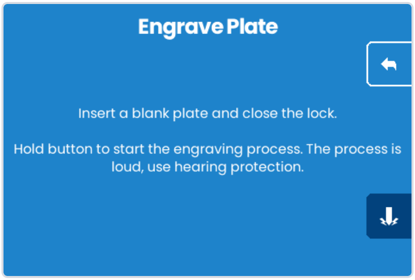
Text entry masks by default

“show” reveals it



Confirm — and it says plainly that nothing here is checked

On the machine — the engrave



Insert a blank plate; hold to start

Countdown while cutting

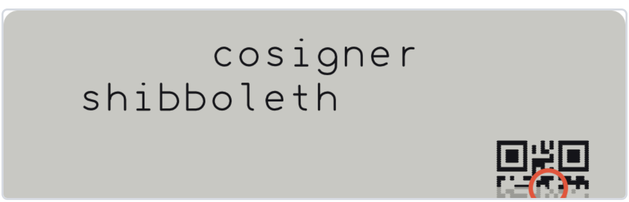
Completed

The plate overlay, live

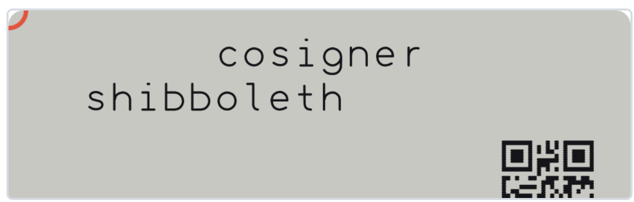
At the moment the engrave begins the emulator draws the **whole planned layout in grey**, from the same `PlanEngraving` call the firmware makes. What the driver has actually stepped is filled in **black**, decoded from the real step stream — not from the plan. The red circle is the head.

head 45.5, 7.4 mm — mid-title; the whole layout is already drawn in grey

head 14.5, 10.1 mm — “sh” cut, the rest still grey



head 74.4, 24.1 mm — into the QR



complete — head parked at 0.0, 0.0

Why the last frame reads 0.0, 0.0.
The job ends by homing, so plan and progress stay in one coordinate space. Before F-121 was fixed the emulator did not home while the device did, and a resumed cut drew its progress roughly 68 mm away from its own plan.

What this run turned up

1. The emulator freezes when an NFC tag is presented to a gathering flow.

`bundleGatherFlow` samples `Platform.NFCReader()` once on entry, so a tag must already be pending or no scanner goroutine starts at all — that part is merely awkward. The defect is what happens when one *is* pending: the emulator's source is one-shot, so after the single record is read the reader sits at EOF, and the scan loop only sleeps on `scanFailed`. On `io.EOF` it spins. Under Go/wasm, which is cooperatively scheduled on the browser's single thread, a goroutine that never blocks starves the event loop and the page stops responding — confirmed here: after presenting a card, even navigating away from the tab timed out.

The consequence is that the NFC path added for the systemwide-payloads work — the one path spec §8.2 exists to make walkable — cannot currently be walked in the emulator, and a 25-card bundle could not be delivered over it in any case, since one flow entry can consume exactly one tag.

2. CORPUS.md's C6 is still a placeholder, and the 12-key policy is no longer chunk-forcing.

Measured: 13 bytes of bytecode, one md1 string. Whatever the corpus wants C6 to demonstrate, this policy no longer demonstrates it — though it remains an excellent *operator* stress case at 26 plates.

3. `me bundle --preview` requires an empty output directory

and exits 2 otherwise. Correct behaviour — worth knowing before a re-run.

Reproducing this document

```
cd <this directory>
bash transcript.sh > out/transcript.txt 2>&1     # every CLI block
python3 build_pdf.py                              # this PDF

# the cosigner set (run inside the seedhammer flake)
go run ./cmd/journeykeys > keys.json
```

The emulator frames were captured by POSTing `canvas.toDataURL()` to a local receiver, so each screenshot is the device framebuffer exactly — no browser chrome, no scaling.